

ML

Lec 07: Naïve Bayes

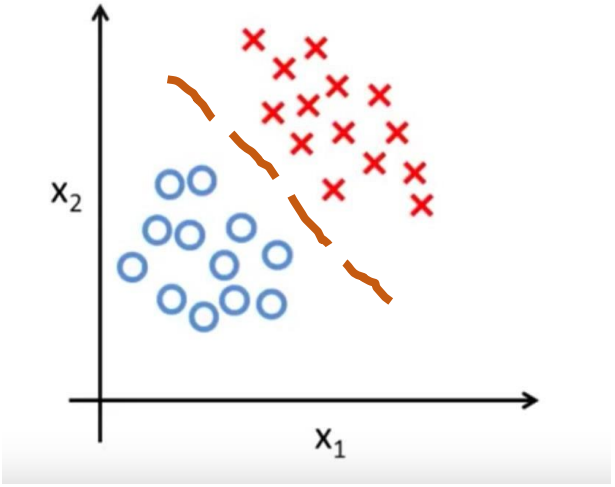
We are about to predict the future!



Some of you will get an A in this course

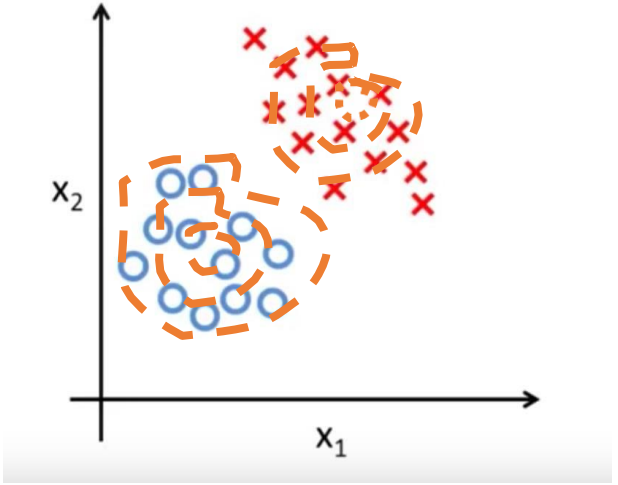
and some of you will not!

Discriminative vs Generative Models



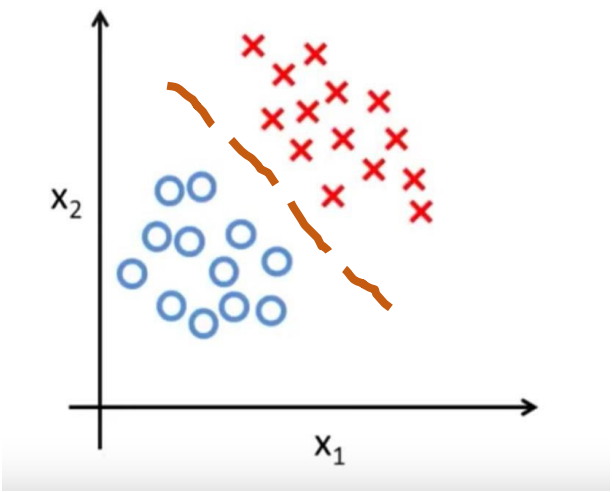
Build a classifier that learns a decision boundary

Classify test data by seeing which side of the boundary it falls into. (what are some examples we've seen?)



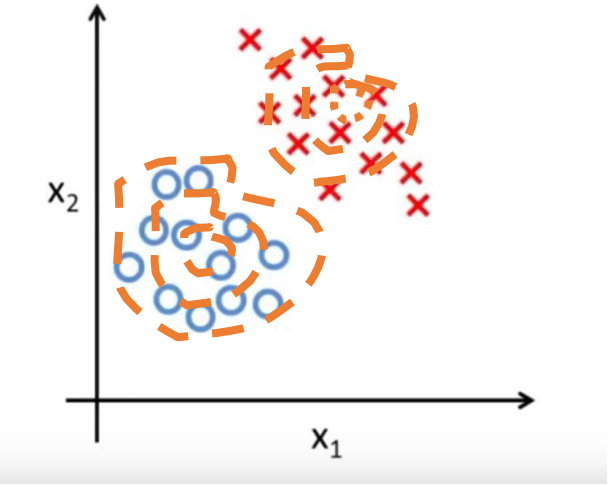
Build a model for what the data in each class looks like (i.e., the distribution of the data)
Classify test data by comparing it against the models for the data in the two classes.

Discriminative vs Generative Models



Build a classifier that learns a decision boundary

Models $p\ (\ y\ |\ x\ )$ directly

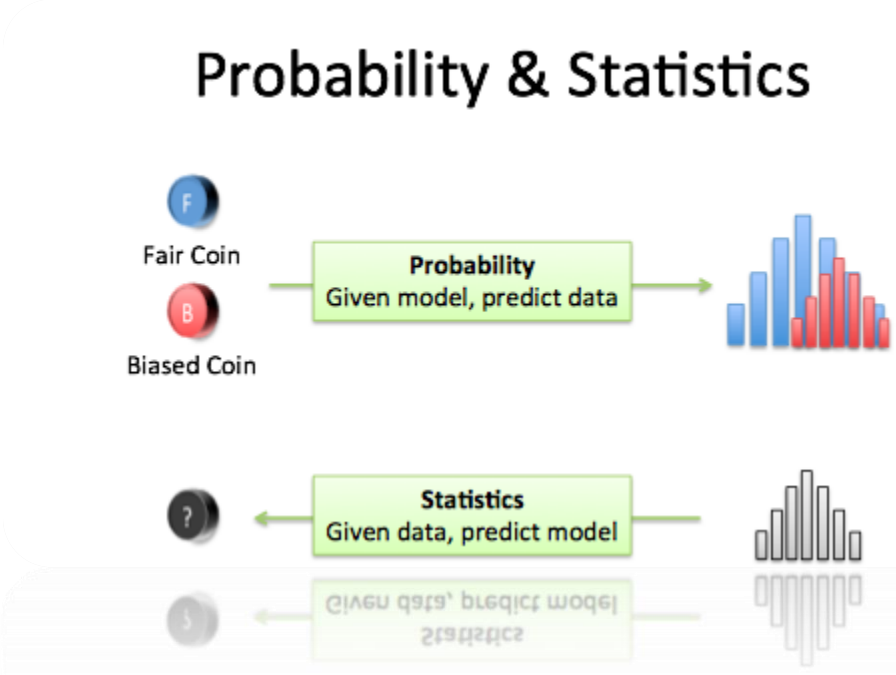


Build a model for what the data in each class looks like (i.e., the distribution of the data)

Models $p\ (\ x\ |\ y\ )$ for each value of y and $p\ (\ y)$ and then use Bayes' rule to find $p\ (\ y\ |\ x\ )$

Probability and Statistics

- **Probability** is starting with an animal, and figuring out what footprints it will make.
- **Statistics** is seeing a footprint, and guessing the animal.
- Probability is straightforward: you have the bear. Measure the foot size, the leg length, and you can deduce the footprints. "Oh, Mr. Bubbles weighs 400lbs and has 3-foot legs, and will make tracks like this." More academically: "We have a fair coin. After 10 flips, here are the possible outcomes."
- Statistics is harder. We measure the footprints and have to guess what animal it could be. A bear? A human? If we get 6 heads and 4 tails, what're the chances of a fair coin?



(https://betterexplained.com/articles/a-brief-introduction-to-probability-statistics/)

Probability

Fair coin toss:

- Probability of heads: ½ ← All possible outcomes in the current universe
- Probability of tails: ½

Fair dice roll

- Probability of getting a 6: 1/6 ← All possible outcomes in the current universe

Probability-Joint and Conditional Probability

Suppose we throw a white and black dice simultaneously. What is the probability that the outcomes would sum to 3?

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- Only one possibility (1) out of 6 possible outcomes remain
- So: P(sums to 3|already a 2 on black dice) = 1/6 (conditional probability)

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- Only one possibility (1) out of 6 possible outcomes remain
- So: P(sums to 3|already a 2 on black dice) = 1/6 (conditional probability)
- P(E|F) = P(E AND F) / P(F) where, E=White dice rolls 1, F=Black dice rolls 2

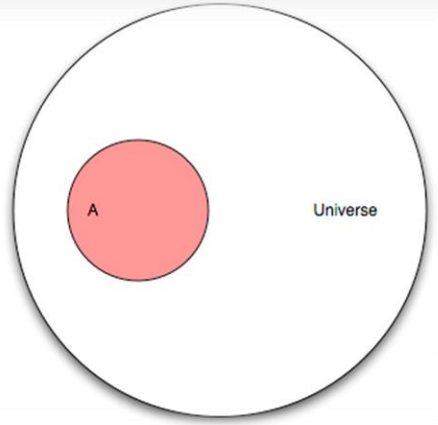
  = (1/36)/(1/6) = 1/6

From Probability to Bayes Theorem

- A *Universe* with all the possible outcomes
- Interested in some subset of them (some *event*)
- Say we are studying cancer:
  - We observe people and see whether they have cancer or not.
  - If we take as our Universe all people participating in our study, then there are two possible outcomes for any particular individual: either he has cancer or not.
  - We can then split our universe in two events:
    - the event “people with cancer” (designated as A),
    - the event “people with no cancer” (or ¬A)

(https://oscarbonilla.com/2009/05/visualizing-bayes-theorem/)

An Event



- So what is the probability that a randomly chosen person has cancer?
- It is just the number of elements in A divided by the number of elements of U (the Universe).
- We denote the number of elements of A as |A|, and read it the *cardinality of* A. And define the probability of A, P(A), as

$$P(A) = \frac{|A|}{|U|}$$

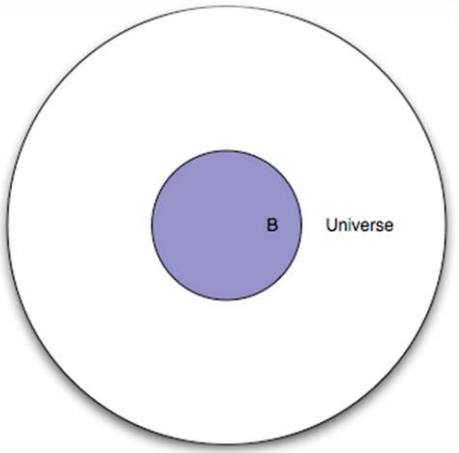
- Since A can have at most the same number of elements as U, the probability P(A) can be at most one.

(https://oscarbonilla.com/2009/05/visualizing-bayes-theorem/)

Another Event

- Let's say there is a new screening test that is supposed to measure something.
- That test will be "positive" for some people, and "negative" for some other people.
- If we take the event B to mean "people for which the test is positive".
- So what is the probability that the test will be "positive" for a randomly selected person?

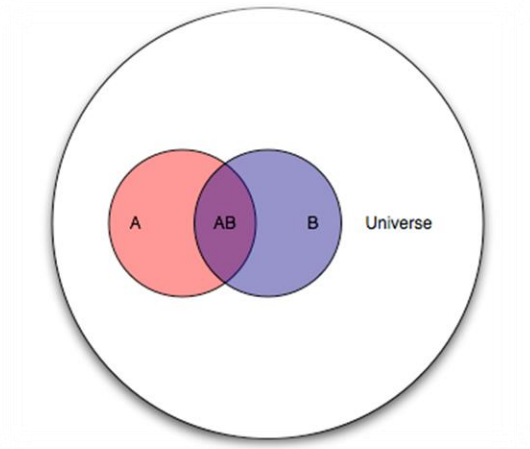
$$P(B) = \frac{|B|}{|U|}$$



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The Two Events Jointly

So far, we have treated the two events in isolation. What happens if we put them together?



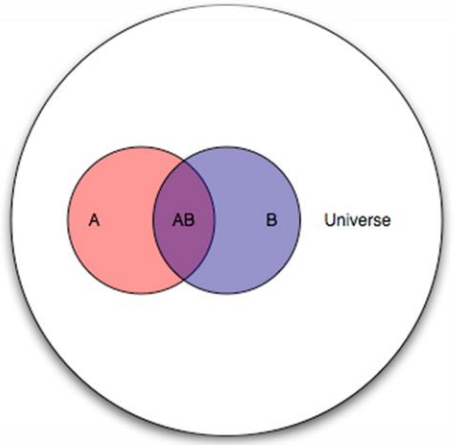
- We can compute the probability of both events occurring (AB is a shorthand for A∩B) in the same way.

$$P(AB) = \frac{|AB|}{|U|}$$

https://oscarbonilla.com/2009/05/visualizing-bayes-theorem/

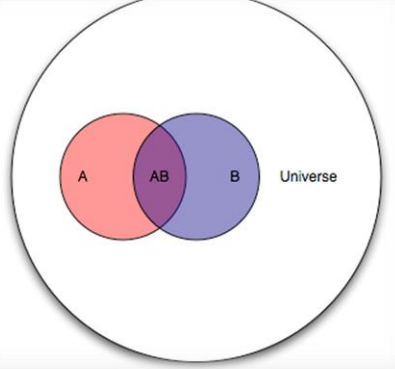
The Two Events Jointly

- We are dealing with:
  - an entire Universe (all people),
  - the event A (people with cancer),
  - the event B (people for whom the test is positive)
- There is also an overlap, the event AB:
  - “people with cancer and with a positive test result”.
- There is also the event B−AB:
  - “people without cancer and with a positive test result”,
- And the event A−AB:
  - “people with cancer and with a negative test result”.



https://oscarbonilla.com/2009/05/visualizing-bayes-theorem/

Conditional Probability



- Now, the question:
  - “given that the test is positive for a randomly selected individual (event B), what is the probability that said individual has cancer?”
- In terms of our Venn diagram:
  - “given that we are in region B, what is the probability that we are in region AB?”

  or stated another way:
  - “if we make region B our new Universe, what is the probability of A?”.
  - The notation for this is P(A|B) and it is read “the probability of A given B”.

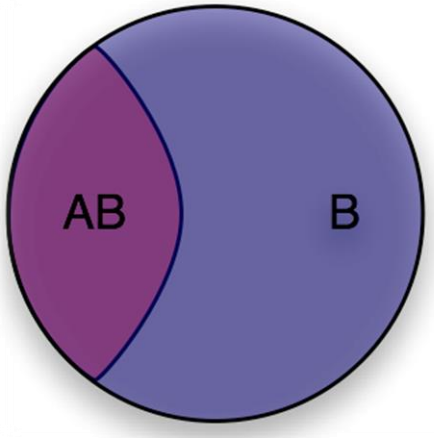
$$P(A|B)=\frac{|AB|}{|B|}$$

Conditional Probability

- And if we divide both the numerator and the denominator by |U|

$$P(A|B) = \frac{\frac{|AB|}{|U|}}{\frac{|B|}{|U|}} = \frac{P(AB)}{P(B)}$$

- What we've effectively done is change the Universe from U (all people), to B (people for whom the test is positive), but we are still dealing with probabilities defined in U.



The Bayes Theorem

$$P(A|B) = \frac{P(B|A)\,P(A)}{P(B)}$$

- Now let's ask the converse question:
  - "given that a randomly selected individual has cancer (event A), what is the probability that the test is positive for that individual (event AB)?".
- It's easy to see that it is:

$$P(B|A) = \frac{P(AB)}{P(A)}$$

- Now we have everything we need to derive Bayes' theorem, putting those two equations together we get

$$P(A|B)P(B) = P(B|A)P(A)$$

- which is to say P(AB) is the same whether you're looking at it from the point of view of A or B, and finally

Bayes Theorem

- Given a hypothesis *h* and Data *D* which bears on the hypothesis:

$$P(h|D) = \frac{p(D|h)\,p(h)}{p(D)}$$

- P(h): Independent probability of h, Class Prior Probability
- P(D): independent probability of D, Predictor Prior Probability
- P(D|h): Conditional Probability of D given h, Likelihood
- P(h |D): Conditional Probability of h given D, Posterior Probability

Bayes: Example 1

Example:

1% of women at age forty who participate in routine screening have breast cancer. 80% of women with breast cancer will get positive mammograms. 9.6% of women without breast cancer will also get positive mammograms.

Bayes: Example 1

	Cancer (1%)	No Cancer (99%)
Test Pos	80%	9.6%
Test Neg	20%	90.4%

- 1% of women have cancer
- If you already have cancer, you are in the first column. There's an 80% chance you will test positive. There's a 20% chance you will test negative.
- If you don't have cancer, you are in the second column. There's a 9.6% chance you will test positive, and a 90.4% chance you will test negative.

Bayes: Example 1

Question:

- A woman in this age group had a positive mammography in a routine screening. What is the probability that she actually has breast cancer?

	Cancer (1%)	No Cancer (99%)
Test Pos	True Pos 1% x 80% = .008	False Pos 99% x 9.6% = .09504
Test Neg	False Neg 1% x 20% = .002	True Neg 99% x 90.4% = .89496

Bayes: Example 1

Question:

- A woman in this age group had a positive mammography in a routine screening. What is the probability that she actually has breast cancer?

	Cancer (1%)	No Cancer (99%)
Test Pos	True Pos 1% x 80% = .008	False Pos 99% x 9.6% = .09504
Test Neg	False Neg 1% x 20% = .002	True Neg 99% x 90.4% = .89496

- The chance of getting a real, positive result is .008. The chance of getting any type of positive result is the chance of a true positive plus the chance of a false positive (.008 + 0.09504 = .10304).
- So, our chance of cancer is .008/.10304 = 0.0776, or about 7.8%.

Bayes: Example 1

$$\Pr(\mathrm{H}|\mathrm{E}) = \frac{\Pr(\mathrm{E}|\mathrm{H})\,\Pr(\mathrm{H})}{\Pr(\mathrm{E}|\mathrm{H})\,\Pr(\mathrm{H}) + \Pr(\mathrm{E}|\text{not H})\,\Pr(\text{not H})}$$

- Pr(H|E) = Chance of having cancer (H) given a positive test (E). This is what we want to know: How likely is it to have cancer with a positive result? In our case it was 7.8%.
- Pr(E|H) = Chance of a positive test (E) given that you had cancer (H). This is the chance of a true positive, 80% in our case.
- Pr(H) = Chance of having cancer (1%).
- Pr(not H) = Chance of not having cancer (99%).
- Pr(E|not H) = Chance of a positive test (E) given that you didn't have cancer (not H). This is a false positive, 9.6% in our case.

$$\Pr(\mathrm{H}|\mathrm{E}) = \frac{\Pr(\mathrm{E}|\mathrm{H})\,\Pr(\mathrm{H})}{\Pr(\mathrm{E})}$$

Example 2:

Day	Outlook	Temperature	Humidity	Wind	Play Tennis
Day1	Sunny	Hot	High	Weak	No
Day2	Sunny	Hot	High	Strong	No
Day3	Overcast	Hot	High	Weak	Yes
Day4	Rain	Mild	High	Weak	Yes
Day5	Rain	Cool	Normal	Weak	Yes
Day6	Rain	Cool	Normal	Strong	No
Day7	Overcast	Cool	Normal	Strong	Yes
Day8	Sunny	Mild	High	Weak	No
Day9	Sunny	Cool	Normal	Weak	Yes
Day10	Rain	Mild	Normal	Weak	Yes
Day11	Sunny	Mild	Normal	Strong	Yes
Day12	Overcast	Mild	High	Strong	Yes
Day13	Overcast	Hot	Normal	Weak	Yes
Day14	Rain	Mild	High	Strong	No

Question:
For the day (sunny, cool, high, strong), what is the play prediction?

Example 2: Multinomial Naïve Bayes

$$P(PlayTennis = yes) = 9/14 = 0.64$$

$$P(PlayTennis = no) = 5/14 = 0.36$$

$$P(Wind = strong \mid PlayTennis = yes) = 3/9 = 0.33$$

$$P(Wind = strong \mid PlayTennis = no) = 3/5 = 0.60$$

etc.

$$P(yes)P(sunny \mid yes)P(cool \mid yes)P(high \mid yes)P(strong \mid yes) = 0.0053$$

$$P(no)P(sunny \mid no)P(cool \mid no)P(high \mid no)P(strong \mid no) = \mathbf{0.0206}$$

$$\Rightarrow answer : PlayTennis(x) = no$$

Naïve Bayes

- Pros:
  - Fast to train (single pass through data)
  - Fast to test
  - Less overfitting
  - Easy to add/remove classes
  - Handles partial data
- Cons:
  - When naïve i.i.d. assumption does not hold, can perform much worse

Practice Question

A patient takes a lab test and the result comes back positive. It is known that the test returns a correct positive result in only 99% of the cases and a correct negative result in only 95% of the cases. Furthermore, only 0.03 of the entire population has this disease.

1. What is the probability that this patient has cancer?
2. What is the probability that he does not have cancer?
3. What is the diagnosis?

Spam Filtering Example : Bernoulli Naïve Bayes

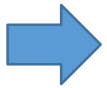
Email	Spam?
send us your password	Y
send us your review	N
review your password	N
review us	Y
send your password	Y
send us your account	Y
review us now	????

- Observe attributes $x_1, x_2, .., x_n$ (e.g., keyword 1, 2, 3, .. are present in the email, respectively)
- Goal: classify email as **spam** or **non-spam**

"Bag of words" features: features based on the appearance of keywords in the text, disregarding the order of the words

Naïve Bayes Spam Filter

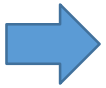
Email	Spam?
send us your password	Y
send us your review	N
review your password	N
review us	Y
send your password	Y
send us your account	Y
review us now	????



	p(spam) = 4/6	p(notspam) = 2/6
	p(w|spam)	p(w|notspam)
password	2/4	1/2
review	1/4	2/2
send	3/4	1/2
us	3/4	1/2
your	3/4	1/2
account	1/4	0/2

Naïve Bayes Spam Filter

Email	Spam?
send us your password	Y
send us your review	N
review your password	N
review us	Y
send your password	Y
send us your account	Y
review us ~~now~~	????



	p(spam) = 4/6	p(notspam) = 2/6
	p(w|spam)	p(w|notspam)
password	2/4	1/2
review	1/4	2/2
send	3/4	1/2
us	3/4	1/2
your	3/4	1/2
account	1/4	0/2

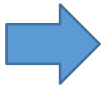
p(review us | spam) = (1-2/4)(1/4)(1-3/4)(3/4)(1-3/4)(1-1/4)

$$p(spam \mid review\ us) = \frac{p(review\ us \mid spam)p(spam)}{p(review\ us \mid spam)p(spam) + p(review\ us \mid notspam)p(notspam)}$$

$$p(not\ spam \mid review\ us) = \frac{p(review\ us \mid notspam)p(notspam)}{p(review\ us \mid spam)p(spam) + p(review\ us \mid notspam)p(notspam)}$$

Naïve Bayes Spam Filter

Email	Spam?
send us your password	Y
send us your review	N
review your password	N
review us	Y
send your password	Y
send us your account	Y
review us ~~now~~	????



	p(spam) = 4/6	p(notspam) = 2/6
	p(w|spam)	p(w|notspam)
password	2/4	1/2
review	1/4	2/2
send	3/4	1/2
us	3/4	1/2
your	3/4	1/2
account	1/4	0/2

p(review us | spam) = (1-2/4)(1/4)(1-3/4)(3/4)(1-3/4)(1-1/4)
p(review us | notspam) = (1-1/2)(2/2)(1-1/2)(1/2)(1-1/2)(1-0/2)

$$p(spam \mid review\ us) = \frac{p(review\ us \mid spam)p(spam)}{p(review\ us \mid spam)p(spam) + p(review\ us \mid notspam)p(notspam)}$$

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